



INTRODUCTION TO PROBABILITY AND STATISTICS FOURTEENTH EDITION

Chapter 10 Inference from Small Samples

INTRODUCTION



- When the sample size is small, the estimation and testing procedures of Chapter 8 are not appropriate.
- There are equivalent small sample test and estimation procedures for
 - ✓ μ , the mean of a normal population
 - ✓ $\mu_1 - \mu_2$, the difference between two population means
 - ✓ σ^2 , the variance of a normal population
 - ✓ The ratio of two population variances, σ_1^2 / σ_2^2 .



GENERAL PRINCIPLE

Once we observe data: (1) we specify a model that can describe the data; (2) we focus on model parameters.

For a dataset, we do the following:

1. Model specification
2. Point estimator(s)
3. Standardize the point estimator and identify its distribution
 1. Find the Confidence Interval
 2. Find the test statistics in a Hypothesis Test



MODELS WE HAVE IN THIS COURSE

- No assumption on the model, but
 - n is large: ch8, 9 **Central Limit Theorem**
 - $X \sim \mathcal{G}(\mu, \sigma^2)$ for some distribution $\mathcal{G}$
 - $X \sim \text{Bernoulli}(p)$
 - n is small: ch15 Nonparametric statistics
- n is small but $X \sim N(\mu, \sigma^2)$: ch 10



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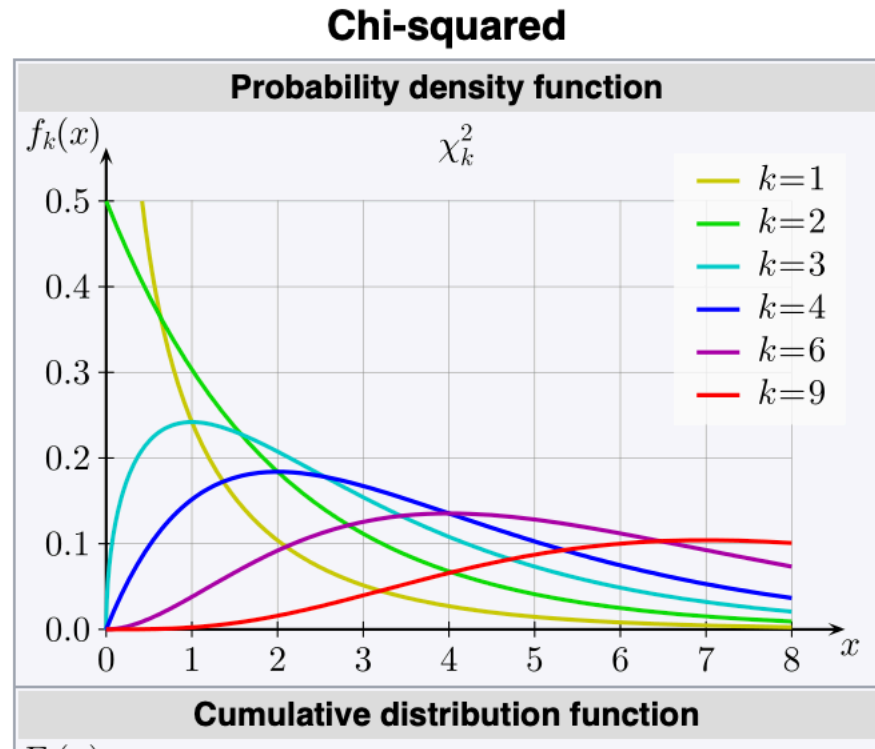
10.1 STUDENT'S t DISTRIBUTION

CHI-SQUARED DISTRIBUTION

If $Z_i \stackrel{i.i.d.}{\sim} N(0,1)$ for $i = 1, \dots, m$, then

$$\sum_{i=1}^m Z_i^2 \sim \chi_m^2.$$

χ_m^2 is called the chi-squared distribution with degrees of freedom m .

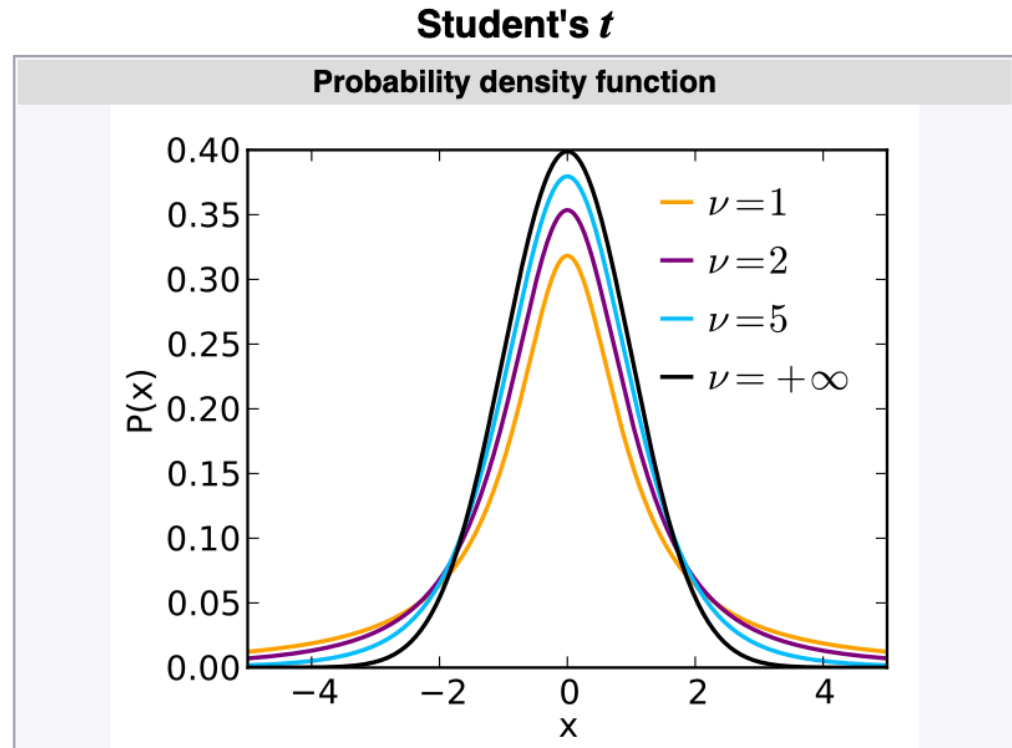


STUDENT'S t DISTRIBUTION

If $U \sim \chi_m^2$,
 $Z \sim N(0,1)$, and U
and Z are
independent, then

$$\frac{Z}{\sqrt{U/m}} \sim t_m.$$

t_m is called the t
distribution with
degrees of
freedom m .



FEATURES OF STUDENT'S t DISTRIBUTION

1. Symmetric and mounded shaped.
2. When degrees of freedom goes to infinity, t_m approaches the standard normal distribution.



Statistician William Sealy
Gosset, known as "Student" 

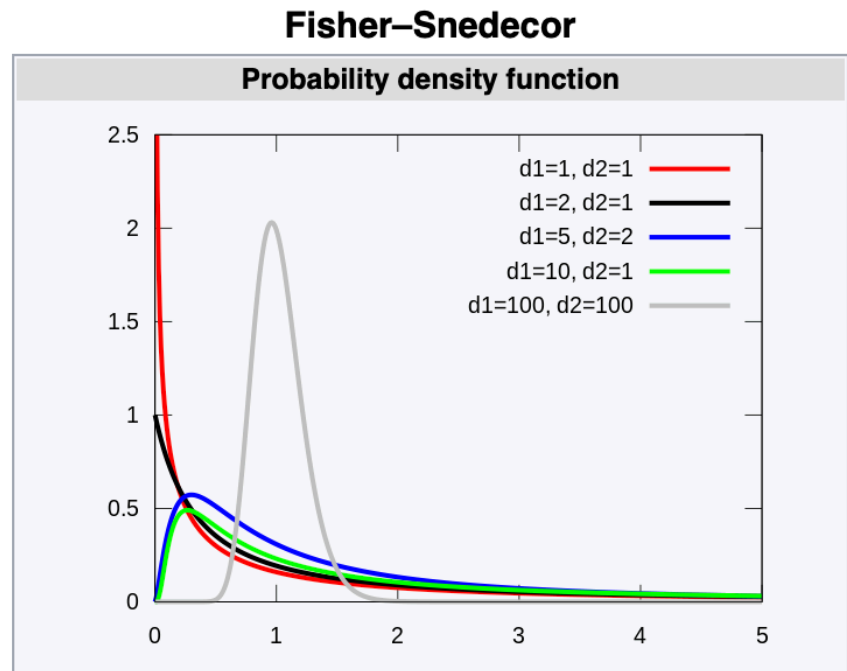
[https://en.wikipedia.org/wiki/
Student's t-distribution](https://en.wikipedia.org/wiki/Student's_t-distribution)

F DISTRIBUTION

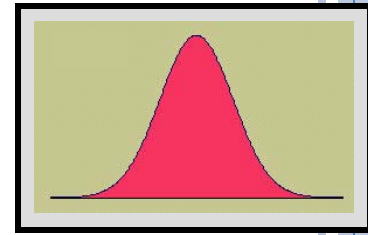
If $U \sim \chi_m^2$, $V \sim \chi_n^2$, and U and V are independent, then

$$\frac{U/m}{V/n} \sim F_{m,n}.$$

$F_{m,n}$ is called the F distribution with degrees of freedom m and n .



THE SAMPLING DISTRIBUTION OF THE SAMPLE MEAN



- When we take a sample from a normal population, the sample mean $\bar{X}$ has a normal distribution for any sample size n , and

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

has a standard normal distribution.

- But if σ is unknown, and we must use S to estimate it, the resulting statistic **is not normal**. $\frac{\bar{X} - \mu}{S/\sqrt{n}}$ is not normal!

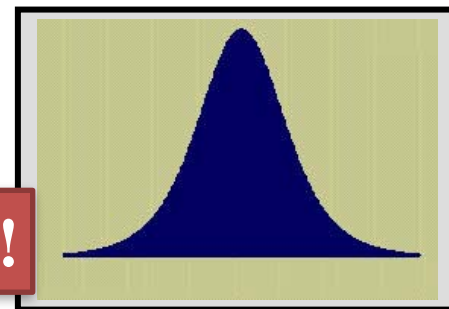


STUDENT'S T DISTRIBUTION

- Fortunately, this statistic does have a sampling distribution that is well known to statisticians, called the **Student's t distribution**, with **$n-1$** degrees of

freedom. $t = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{(n-1)}$

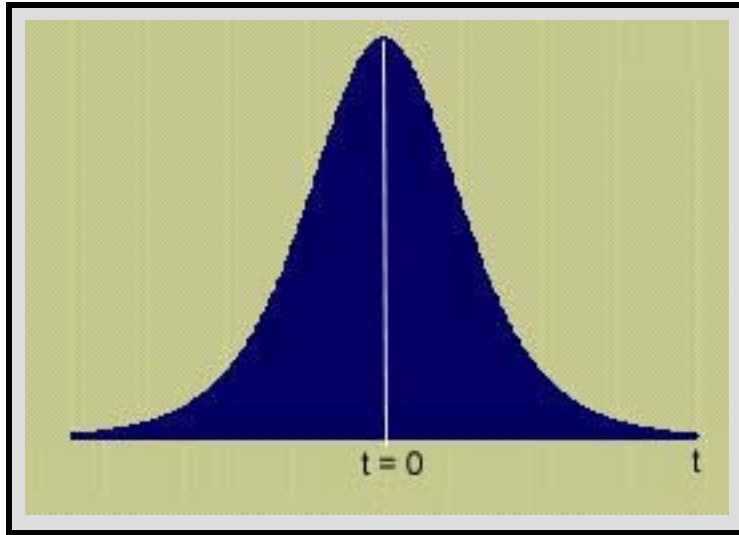
Why? Go to Math Stat!



- We can use this distribution to create estimation testing procedures for the population mean μ .

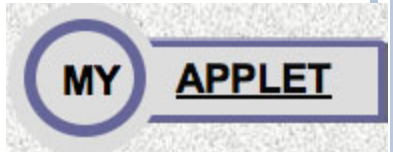


PROPERTIES OF STUDENT'S t



- **Mound-shaped** and symmetric about 0.
- **More variable than z** , with “heavier tails”

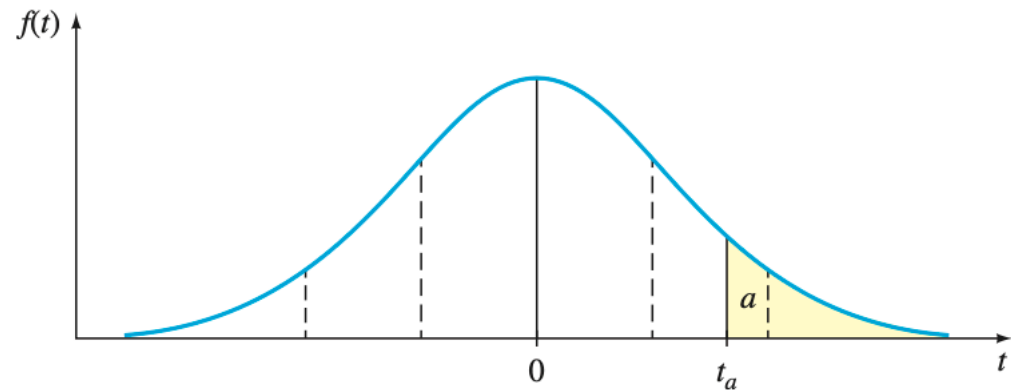
- Shape depends on the sample size n or the **degrees of freedom, $n-1$** .
- As n increases the shapes of the t and z distributions become almost identical.



STUDENT'S t TABLE

- I use $t_{\nu;\alpha}$ to denote critical point of the distribution for the right-tailed probability of α .

- $P(t_{\nu} > t_{\nu;\alpha}) = \alpha$.



■ Table 10.1 Format of the Student's t Table from Table 4 in Appendix I

df	$t_{.100}$	$t_{.050}$	$t_{.025}$	$t_{.010}$	$t_{.005}$	df
1	3.078	6.314	12.706	31.821	63.657	1
2	1.886	2.920	4.303	6.965	9.925	2
3	1.638	2.353	3.182	4.541	5.841	3
4	1.533	2.132	2.776	3.747	4.604	4
5	1.476	2.015	2.571	3.365	4.032	5
6	1.440	1.943	2.447	3.143	3.707	6
7	1.415	1.895	2.365	2.998	3.499	7
8	1.397	1.860	2.306	2.896	3.355	8
9	1.383	1.833	2.262	2.821	3.250	9
·	·	·	·	·	·	·
·	·	·	·	·	·	·
·	·	·	·	·	·	·
28	1.313	1.701	2.048	2.467	2.763	28
29	1.311	1.699	2.045	2.462	2.756	29
30	1.310	1.697	2.042	2.457	2.750	30
·	·	·	·	·	·	·
·	·	·	·	·	·	·
·	·	·	·	·	·	·
100	1.290	1.660	1.984	2.364	2.626	100
200	1.286	1.653	1.972	2.345	2.601	200
300	1.284	1.650	1.968	2.339	2.592	300
400	1.284	1.649	1.966	2.336	2.588	400
500	1.283	1.648	1.965	2.334	2.586	500
inf.	1.282	1.645	1.96	2.326	2.576	inf.

USING THE *t*-TABLE

- Table 4 gives the values of t that cut off certain critical values in the tail of the t distribution.
- Index df and the appropriate tail area α to find t_{α} , the value of t with area α to its right.

df	$t_{.100}$	$t_{.050}$	$t_{.025}$	$t_{.010}$
1	3.078	6.314	12.706	31.821
2	1.886	2.920	4.303	6.965
3	1.638	2.353	3.182	4.541
4	1.533	2.132	2.776	3.747
5	1.476	2.015	2.571	3.365
6	1.440	1.943	2.447	3.143
7	1.415	1.895	2.365	2.998
8	1.397	1.860	2.306	2.896
9	1.383	1.833	2.262	2.821
10	1.372	1.812	2.228	2.764
11	1.363	1.796	2.201	2.718
12	1.356	1.782	2.179	2.681
13	1.350	1.771	2.160	2.650
14	1.345	1.761	2.145	2.624
15	1.341	1.753	2.131	2.602

For a random sample of size $n = 10$, find a value of t that cuts off .025 in the right tail.

Row = $df = n - 1 = 9$

Column subscript = $\alpha = .025$

$$t_{.025} = 2.262$$

RECALL CH 8 & 9

Suppose $X_i \stackrel{i.i.d.}{\sim} \mathcal{G}(\mu, \sigma^2)$ for $i = 1, \dots, n$.

(1) When σ is known, the Central Limit Theorem:

$$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d.} N(0,1) \text{ as } n \rightarrow \infty$$

(2) When σ is unknown, Slutsky's theorem:

$$\frac{\bar{X}_n - \mu}{S/\sqrt{n}} \xrightarrow{d.} N(0,1) \text{ as } n \rightarrow \infty$$

$\xrightarrow{d.}$

- Asymptotically distributed as 漸進分配
- Converges in distribution 分配收敛

CHAPTER 10

Suppose $X_i \stackrel{i.i.d.}{\sim} N(\mu, \sigma^2)$ for $i = 1, \dots, n$.

(1) When σ is known,

$$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \sim N(0,1).$$

(2) When σ is unknown,

$$\frac{\bar{X}_n - \mu}{S/\sqrt{n}} \sim t_{n-1}.$$



The left side of the slide features a series of vertical stripes in various shades of blue, ranging from light to dark. Overlaid on these stripes are several circles of different sizes, also in shades of blue, creating a modern, abstract design.

10.2 SMALL-SAMPLE INFERENCES CONCERNING A POPULATION MEAN

EXAMPLE



A sprinkler system is designed so that the average time for the sprinklers to activate after being turned on is no more than 15 seconds. A test of 6 systems gave the following times:

17, 31, 12, 17, 13, 25

Is the system working as specified? Test using $\alpha = .05$.

$H_0 : \mu = 15$ (working as specified)

$H_a : \mu > 15$ (not working as specified)



KEY INGREDIENTS (WHEN σ IS UNKNOWN)

1. Model: $X_i \overset{i.i.d.}{\sim} N(\mu, \sigma^2)$ and we focus on the estimation of μ .
2. Point estimator of μ : $\bar{X}$
3. If σ^2 is unknown, we use the sample variance S^2 to estimate it. Then, the *standardized point* estimator and its distribution: $\frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{(n-1)}.$



CONFIDENCE INTERVAL

$$1 - \alpha = P(-t_{(n-1);\alpha/2} < \frac{\bar{X} - \mu}{S/\sqrt{n}} < t_{(n-1);\alpha/2})$$

$$= P(-t_{(n-1);\alpha/2} \frac{S}{\sqrt{n}} < \bar{X} - \mu < t_{(n-1);\alpha/2} \frac{S}{\sqrt{n}})$$

$$= P(-t_{(n-1);\alpha/2} \frac{S}{\sqrt{n}} < \mu - \bar{X} < t_{(n-1);\alpha/2} \frac{S}{\sqrt{n}})$$

$$= P(\bar{X} - t_{(n-1);\alpha/2} \frac{S}{\sqrt{n}} < \mu < \bar{X} + t_{(n-1);\alpha/2} \frac{S}{\sqrt{n}})$$

- With data, the $100(1 - \alpha) \%$ CI for the population mean μ as $\bar{X} \pm t_{(n-1);\alpha/2} \frac{S}{\sqrt{n}}$.



HYPOTHESIS TEST

1. $H_0 : \mu = \mu_0$ versus $H_a : \mu \neq \mu_0$ (or $>$ or $<$)
2. Set the value of α .
3. The test statistic and its sampling distribution is

$$t_{STAT} = \frac{\bar{X} - \mu_0}{S/\sqrt{n}} \sim t_{(n-1)}.$$

4. Calculate the realized statistic, $t^* = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$.

5. Find the rejection region or p -value.

6. Conclude.

- Reject or do not reject H_0 .
- Conclude in plain language and get back to the scenario.



EXAMPLE



Data: 17, 31, 12, 17, 13, 25

First, calculate the sample mean and standard deviation, using your calculator or the formulas in Chapter 2.

$$\bar{x} = \frac{\sum x_i}{n} = \frac{115}{6} = 19.167$$

$$s = \sqrt{\frac{\sum x^2 - \frac{(\sum x)^2}{n}}{n-1}} = \sqrt{\frac{2477 - \frac{115^2}{6}}{5}} = 7.387$$



EXAMPLE



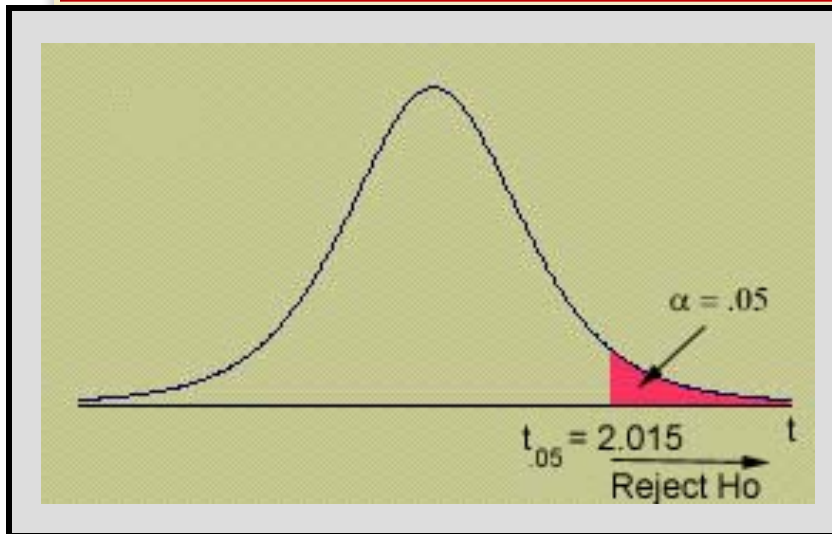
Data: 17, 31, 12, 17, 13, 25

Calculate the test statistic and find the rejection region for $\alpha = .05$.

Test statistic :

Degrees of freedom :

$$t = \frac{\bar{x} - \mu_0}{s / \sqrt{n}} = \frac{19.167 - 15}{7.387 / \sqrt{6}} = 1.38 \quad df = n - 1 = 6 - 1 = 5$$



Rejection Region: Reject H_0 if $t > 2.015$. If the test statistic falls in the rejection region, its p -value will be less than $\alpha = .05$.

CONCLUSION



Data: 17, 31, 12, 17, 13, 25

Compare the observed test statistic to the rejection region, and draw conclusions.

$$H_0 : \mu = 15$$

$$H_a : \mu > 15$$

Test statistic : $t = 1.38$

Rejection Region :

Reject H_0 if $t > 2.015$.

Conclusion: For our example, $t = 1.38$ does not fall in the rejection region and H_0 is not rejected. There is insufficient evidence to indicate that the average activation time is greater than 15.

STATISTICAL CONCLUSION (CAREFUL!)

- Because realized statistic does not fall into the rejection region, so we do not reject H_0 .
- Do not reject H_0
- Insignificance evidence to show H_0 is incorrect $\approx$
Insignificant evidence to show that H_a is correct



SOP WITH REJECTION REGION APPROACH

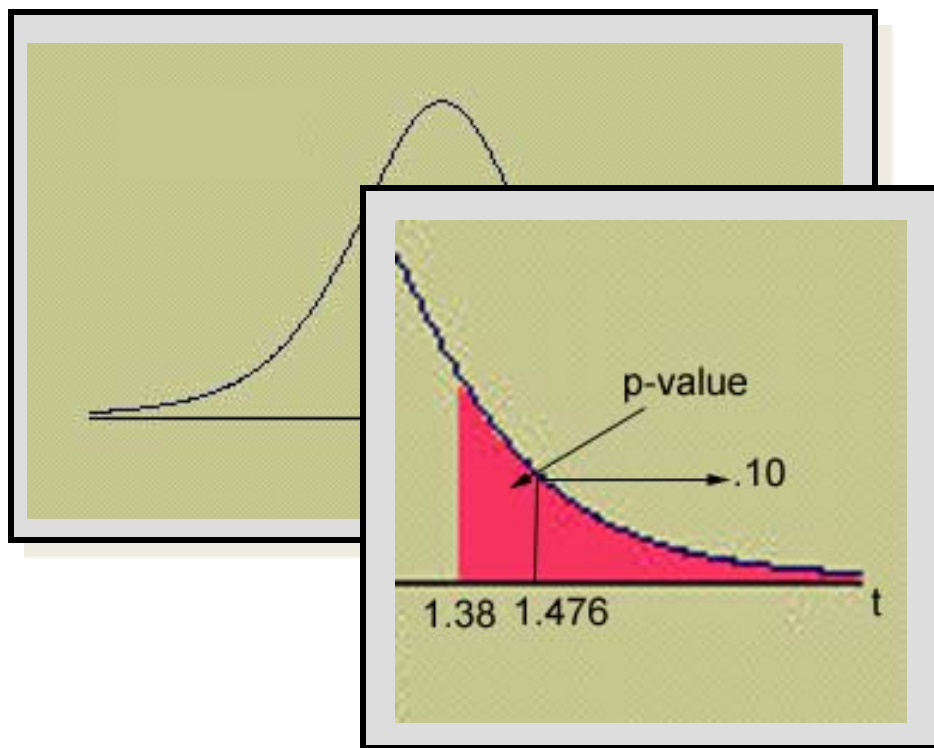
1. $H_0 : \mu = 15$ versus $H_a : \mu > 15$
2. Set $\alpha = 0.05$.
3. Define the test statistic $t_{STAST} = \frac{\bar{X} - 15}{S/\sqrt{n}} \sim t_{(6-1)}$.
4. Calculate the realized test statistic
 $t^* = 1.38$.
5. Rejection region $\{t : t > 2.015\}$.
6. Because t^* does not fall into the rejection region, we do not reject H_0 .



APPROXIMATING THE *P*-VALUE



- You can only approximate the *p*-value for the test using Table 4.



<i>df</i>	$t_{.100}$	$t_{.050}$
1	3.078	6.314
2	1.886	2.920
3	1.638	2.353
4	1.533	2.132
5	1.476	2.015

Since the observed value of $t = 1.38$ is smaller than $t_{.10} = 1.476$,

$p\text{-value} > .10$.

p-VALUE APPROACH

5. The p-value $> 0.10 > \alpha = 0.05$
6. We do not rejection H_0 .

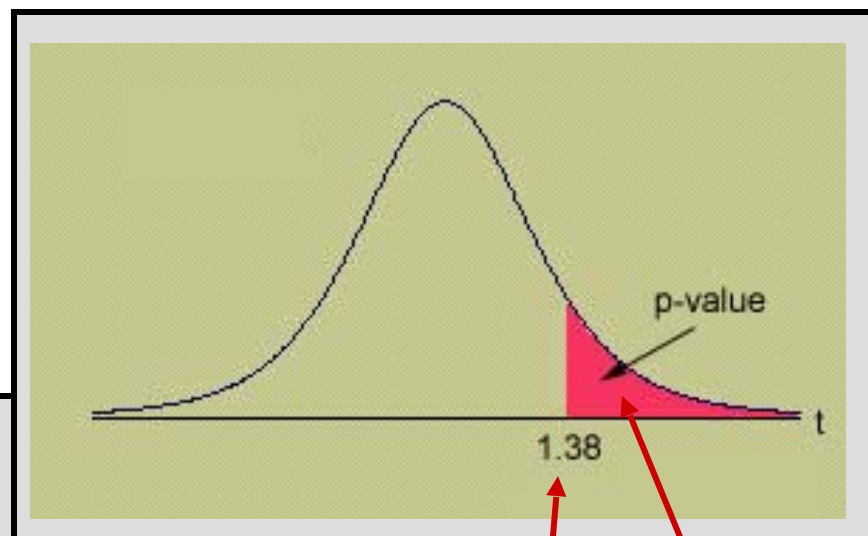




THE EXACT P -VALUE

- You can get the exact p -value using some calculators or a computer.

p -value = .113 which is greater than .10 as we approximated using Table 4.



One-Sample T: Times

Test of $\mu = 15$ vs > 15

Variable	N	Mean	StDev	SE Mean	95% Lower Bound
Times	6	19.1667	7.3869	3.0157	13.0899

T	P
1.38	0.113

A decorative graphic on the left side of the slide. It features a series of vertical stripes in various shades of blue and white. Overlaid on these stripes are several circles of different sizes, also in shades of blue. The circles are arranged in a way that they appear to be floating or overlapping the stripes.

10.3

Small sample inference for the difference between
two population means: independent random
sample

EXAMPLE

- Two training procedures are compared by measuring the time that it takes trainees to assemble a device. A different group of trainees are taught using each method. Is there a difference in the two methods? Use $\alpha = .01$.



Time to Assemble	Method 1	Method 2
Sample size	10	12
Sample mean	35	31
Sample Std Dev	4.9	4.5

$$H_0 : \mu_1 - \mu_2 = 0$$

$$H_a : \mu_1 - \mu_2 \neq 0$$

Test statistic :

$$t = \frac{\bar{x}_1 - \bar{x}_2 - 0}{\sqrt{s^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$$

RECALL

- A linear combination of normal distribution remains a normal distribution:

- $(\bar{X}_1 - \bar{X}_2) \sim N\left(\mu_1 - \mu_2, \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}\right)$



KEY INGREDIENTS

1. Model

$$X_{1i} \sim N(\mu_1, \sigma_1^2), \text{ for } i = 1, \dots, n_1$$

$$X_{2j} \sim N(\mu_2, \sigma_2^2), \text{ for } j = 1, \dots, n_2$$

2. Point estimator of $(\mu_1 - \mu_2)$: $(\bar{X}_1 - \bar{X}_2)$.

3. If σ_1^2 and σ_2^2 are known, the normalized point estimator and its sampling distribution:

$$\frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \sim Z.$$



σ_1^2 AND σ_2^2 ARE KNOWN

We estimate them using sample variances S_1^2 and S_2^2 :

$$S_1^2 = \frac{1}{n_1 - 1} \sum_{i=1}^{n_1} (X_{1i} - \bar{X}_1)^2$$

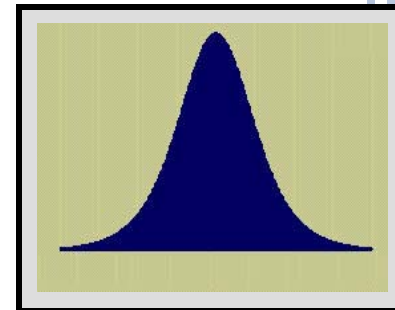
$$S_2^2 = \frac{1}{n_2 - 1} \sum_{j=1}^{n_2} (X_{2j} - \bar{X}_2)^2$$

Then, we furthermore consider two cases:

1. $\sigma_1^2 = \sigma_2^2$
2. $\sigma_1^2 \neq \sigma_2^2$



TESTING THE DIFFERENCE BETWEEN TWO MEANS



- How can you tell if the equal variance assumption is reasonable?

Back in chapter 10.6!

Rule of Thumb :

If the ratio, $\frac{\text{larger } s^2}{\text{smaller } s^2} \leq 3$,

the equal variance assumption is reasonable.

If the ratio, $\frac{\text{larger } s^2}{\text{smaller } s^2} > 3$,

use an alternative test statistic.



CASE I: $\sigma_1^2 = \sigma_2^2$. *POOLED METHOD*

Assume σ_1^2 and σ_2^2 are unknown but they are equal

Define the common sample variance by

$$S^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2},$$

Then, we have

$$\frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{S^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} \sim t_{(n_1+n_2-2)}$$



CI

Follow the same manner:

$$\begin{aligned} 1 - \alpha &= P \left(-t_{(n_1+n_2-2);\alpha/2} < \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{S^2(\frac{1}{n_1} + \frac{1}{n_2})}} < t_{(n_1+n_2-2);\alpha/2} \right) \\ &= P \left(-t_{(n_1+n_2-2);\alpha/2} \sqrt{S^2(\frac{1}{n_1} + \frac{1}{n_2})} < (\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2) < t_{(n_1+n_2-2);\alpha/2} \sqrt{S^2(\frac{1}{n_1} + \frac{1}{n_2})} \right) \\ &= P \left((\bar{X}_1 - \bar{X}_2) - t_{(n_1+n_2-2);\alpha/2} \sqrt{S^2(\frac{1}{n_1} + \frac{1}{n_2})} < (\mu_1 - \mu_2) < (\bar{X}_1 - \bar{X}_2) + t_{(n_1+n_2-2);\alpha/2} \sqrt{S^2(\frac{1}{n_1} + \frac{1}{n_2})} \right). \end{aligned}$$

The $100(1 - \alpha) \%$ CI for $(\mu_1 - \mu_2)$ is

$$(\bar{X}_1 - \bar{X}_2) \pm t_{(n_1+n_2-2);\alpha/2} \sqrt{S^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}.$$



HYPOTHESIS TEST

3. For $H_0 : \mu_1 - \mu_2 = D_0$, the test statistic and its sampling distribution is

$$t_{STAT} = \frac{(\bar{X}_1 - \bar{X}_2) - D_0}{\sqrt{S^2(\frac{1}{n_1} + \frac{1}{n_2})}} \sim t_{(n_1+n_2-2)}.$$

4. The realized statistic is

$$t^* = \frac{(\bar{x}_1 - \bar{x}_2) - D_0}{\sqrt{s^2(\frac{1}{n_1} + \frac{1}{n_2})}}.$$



EXAMPLE

- Solve this problem by approximating the p -value using Table 4.



Time to Assemble	Method 1	Method 2
Sample size	10	12
Sample mean	35	31
Sample Std Dev	4.9	4.5

Calculate:

$$s^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$
$$= \frac{9(4.9^2) + 11(4.5^2)}{20} = 21.942$$

Test statistic:

$$t = \frac{35 - 31}{\sqrt{21.942 \left(\frac{1}{10} + \frac{1}{12} \right)}}$$
$$= 1.99$$

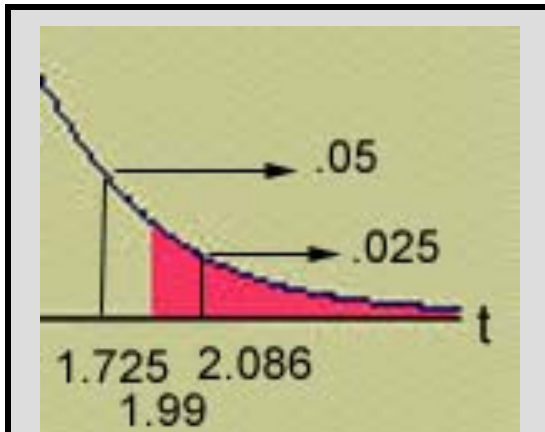
EXAMPLE



$$p\text{-value} : P(t > 1.99) + P(t < -1.99)$$

$$P(t > 1.99) = \frac{1}{2}(p\text{-value})$$

$$df = n_1 + n_2 - 2 = 10 + 12 - 2 = 20$$



$$.025 < \frac{1}{2}(p\text{-value}) < .05$$

$$.05 < p\text{-value} < .10$$

Since the p -value is greater than $\alpha = .01$, H_0 is not rejected. There is insufficient evidence to indicate a difference in the population means.

df	t _{.100}	t _{.050}	t _{.025}	t _{.010}	t _{.005}	df
19	1.328	1.729	2.093	2.539	2.861	19
20	1.325	1.725	2.086	2.528	2.845	20

SOP WITH P-VALUE APPROACH

1. $H_0 : \mu_1 - \mu_2 = 0$ versus $H_a : \mu_1 - \mu_2 \neq 0$.
2. $\alpha = 0.01$.
3. Test statistic and its sampling distribution

$$t_{STAT} = \frac{(\bar{X}_1 - \bar{X}_2) - 0}{\sqrt{S^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} \sim t_{(n_1+n_2-2)=20}.$$

4. Realized test statistic $t^* = 1.99$.
5. Approximate p-value:
 $0.05 = 2 \times 0.025 < p\text{-value} < 2 \times 0.05 = 0.10$.
6. Because $p\text{-value} > 0.01$, we do not reject H_0 .



CASE 2: $\sigma_1^2 \neq \sigma_2^2$. *SATTERTHWAITE'S APPROXIMATION*

Assume σ_1^2 and σ_2^2 are unknown, but $\sigma_1^2 \neq \sigma_2^2$.

3. The standardized point estimator for $(\mu_1 - \mu_2)$ and its sampling distribution is

$$\frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}} \sim t_\nu,$$

with

$$\nu \approx \text{floor} \left(\frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2} \right)^2}{\frac{(s_1^2/n_1)^2}{n_1 - 1} + \frac{(s_2^2/n_2)^2}{n_2 - 1}} \right).$$

We will *round down* to the nearest integer.



CASE 2: $\sigma_1^2 \neq \sigma_2^2$. *SATTERTHWAITE'S APPROXIMATION*

The CI is $(\bar{x}_1 - \bar{x}_2) \pm t_{\nu; \alpha/2} \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$.

For $H_0 : \mu_1 - \mu_2 = D_0$, the test statistic and its sampling distribution is

$$\frac{(\bar{X}_1 - \bar{X}_2) - D_0}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}} \sim t_{\nu}.$$

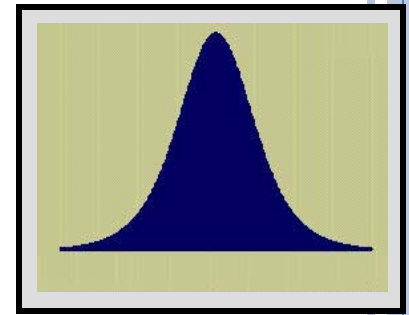


A decorative graphic on the left side of the slide. It features a series of vertical stripes in various shades of blue and white. Overlaid on these stripes are several circles of different sizes, also in shades of blue. The circles are arranged in a way that they appear to be floating or overlapping the stripes.

10.4

**Small-sample inferences for the difference
between two population means: A paired
difference test**

THE PAIRED-DIFFERENCE TEST



- Sometimes the assumption of independent samples is intentionally violated, resulting in a **matched-pairs** or **paired-difference test**.
- By designing the experiment in this way, we can eliminate unwanted variability in the experiment by analyzing only the differences,

$$d_i = x_{1i} - x_{2i}$$

- to see if there is a difference in the two population means, $\mu_1 - \mu_2$.



EXAMPLE

Car	1	2	3	4	5
Type A	10.6	9.8	12.3	9.7	8.8
Type B	10.2	9.4	11.8	9.1	8.3



- One Type A and one Type B tire are randomly assigned to each of the rear wheels of five cars. Compare the average tire wear for types A and B using a test of hypothesis.

$$H_0 : \mu_1 - \mu_2 = 0$$

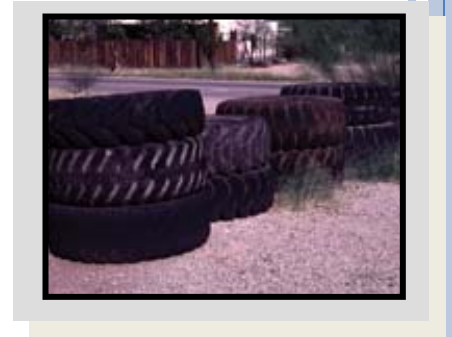
$$H_a : \mu_1 - \mu_2 \neq 0$$

- But the samples are not independent. The pairs of responses are linked because measurements are taken on the same car.



EXAMPLE

Car	1	2	3	4	5
Type A	10.6	9.8	12.3	9.7	8.8
Type B	10.2	9.4	11.8	9.1	8.3



- One Type A and one Type B tire are randomly assigned to each of the rear wheels of five cars. Compare the average tire wear for types A and B using a test of hypothesis.
- But the samples are not independent. The pairs of responses are linked because measurements are taken on the same car.



EXAMPLE



Car	1	2	3	4	5
Type A	10.6	9.8	12.3	9.7	8.8
Type B	10.2	9.4	11.8	9.1	8.3
Difference	.4	.4	.5	.6	.5



KEY INGREDIENTS

1. Model: $X_{1i} - X_{2i} \sim N(\mu_1 - \mu_2, \sigma^2)$, for $i = 1, \dots, n$.

Define $D_i = X_{1i} - X_{2i}$, and $\mu_D = (\mu_1 - \mu_2)$

The model is written as

$$D_i \sim N(\mu_D, \sigma_D^2), \text{ for } i = 1, \dots, n.$$

2. Point estimator for μ_D : $\bar{D} = \frac{1}{n} \sum_{i=1}^n D_i$

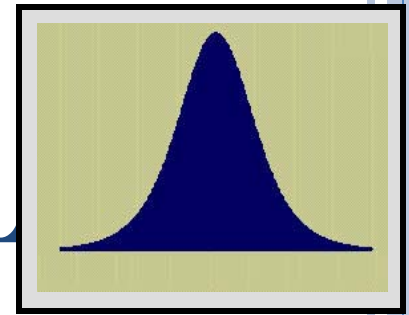
3. Standardized estimator and its distribution:

$$\frac{\bar{D} - (\mu_1 - \mu_2)}{S_D / \sqrt{n}} \sim t_{(n-1)},$$

where $S_D = \sqrt{\frac{\sum_{i=1}^n (D_i - \bar{D})^2}{n-1}}$



CONFIDENCE INTERVAL



Follow the same manner

$$\begin{aligned} 1 - \alpha &= P \left(-t_{(n-1);\alpha/2} < \frac{\bar{D} - (\mu_1 - \mu_2)}{S_D/\sqrt{n}} < t_{(n-1);\alpha/2} \right) \\ &= P \left(-t_{(n-1);\alpha/2} \frac{S_D}{\sqrt{n}} < \bar{D} - (\mu_1 - \mu_2) < t_{(n-1);\alpha/2} \frac{S_D}{\sqrt{n}} \right) \\ &= P \left(\bar{D} - t_{(n-1);\alpha/2} \frac{S_D}{\sqrt{n}} < (\mu_1 - \mu_2) < \bar{D} + t_{(n-1);\alpha/2} \frac{S_D}{\sqrt{n}} \right) \end{aligned}$$

A 100(1- α)% CI for a paired experiment is

$$\bar{d} \pm t_{(n-1),\alpha/2} \frac{s_d}{\sqrt{n}} .$$



HYPOTHESIS TEST

1. $H_0 : \mu_1 - \mu_2 = D_0$
2. Set α
3. the test statistic and its sampling distribution is

$$t_{STAT} = \frac{\bar{D} - D_0}{S_D/\sqrt{n}} \sim t_{(n-1)}$$

4. The realized test statistic $t^* = \frac{\bar{d} - D_0}{s_d/\sqrt{n}}$
5. Rejection rejection or p-value approach
6. Conclude.

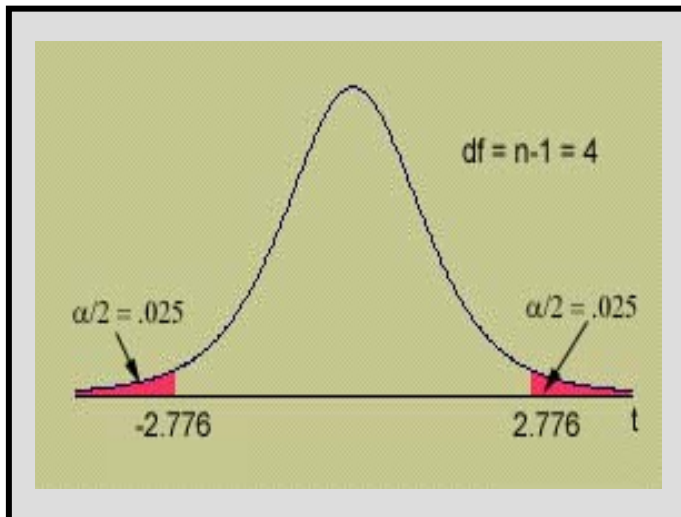


EXAMPLE



Find the 95 % CI for the average difference of tire wear between types A and B!

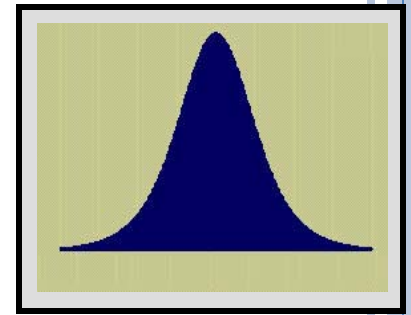
Car	1	2	3	4	5
Type A	10.6	9.8	12.3	9.7	8.8
Type B	10.2	9.4	11.8	9.1	8.3
Difference	.4	.4	.5	.6	.5



$$\text{Calculated } \bar{d} = \frac{\sum d_i}{n} = .48$$

$$s_d = \sqrt{\frac{\sum d_i^2 - \frac{(\sum d_i)^2}{n}}{n-1}} = .0837$$

CI



- Because $\bar{d} = 0.48$, $s_d = 0.0837$, and $t_{4,0.025} = 2.776$, the 95% CI is
$$\bar{d} \pm t_{(n-1);\alpha/2} \frac{s_d}{\sqrt{n}} = \left[0.48 \pm 2.776 \frac{0.0837}{\sqrt{5}} \right] = [0.37, 0.58]$$
- In repeated sampling, about 95 % of intervals constructed this way will cover the true average difference of the tire wear between types A and B.
- There seems a difference of tire wear between types A and B.



EXAMPLE

Car	1	2	3	4	5
Type A	10.6	9.8	12.3	9.7	8.8
Type B	10.2	9.4	11.8	9.1	8.3



- One Type A and one Type B tire are randomly assigned to each of the rear wheels of five cars. Compare the average tire wear for types A and B using a test of hypothesis.

$$H_0 : \mu_1 - \mu_2 = 0$$

$$H_a : \mu_1 - \mu_2 \neq 0$$

- But the samples are not independent. The pairs of responses are linked because measurements are taken on the same car.



THE PAIRED-DIFFERENCE TEST

To test $H_0 : \mu_1 - \mu_2 = 0$ we test $H_0 : \mu_d = 0$
using the test statistic

$$t = \frac{\bar{d} - 0}{s_d / \sqrt{n}}$$

where n = number of pairs, $\bar{d}$ and s_d are the
mean and standard deviation of the differences, d_i .

Use the p - value or a rejection region based on
a t - distribution with $df = n - 1$.

EXAMPLE

Car	1	2	3	4	5
Type A	10.6	9.8	12.3	9.7	8.8
Type B	10.2	9.4	11.8	9.1	8.3
Difference	.4	.4	.5	.6	.5



$$H_0 : \mu_1 - \mu_2 = 0$$

$$H_a : \mu_1 - \mu_2 \neq 0$$

$$\text{Calculated } \bar{d} = \frac{\sum d_i}{n} = .48$$

$$s_d = \sqrt{\frac{\sum d_i^2 - \frac{(\sum d_i)^2}{n}}{n-1}} = .0837$$

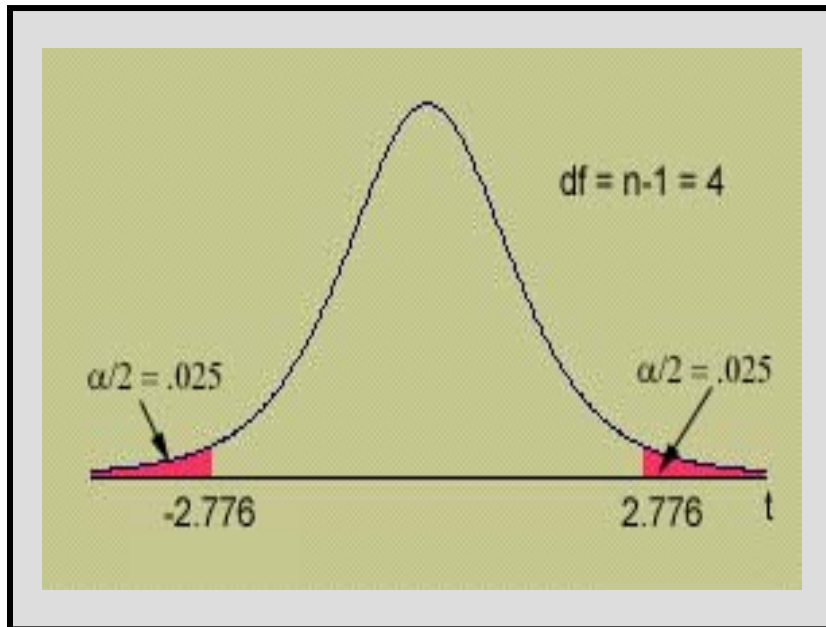
Test statistic :

$$t = \frac{\bar{d} - 0}{s_d / \sqrt{n}} = \frac{.48 - 0}{.0837 / \sqrt{5}} = 12.8$$



EXAMPLE

Car	1	2	3	4	5
Type A	10.6	9.8	12.3	9.7	8.8
Type B	10.2	9.4	11.8	9.1	8.3
Difference	.4	.4	.5	.6	.5



Rejection region: Reject H_0 if $t > 2.776$ or $t < -2.776$.

Conclusion: Since $t = 12.8$, H_0 is rejected. There is a difference in the average tire wear for the two types of tires.

REJECTION REGION APPROACH

1. $H_0 : \mu_1 - \mu_2 = 0$ vs $H_a : \mu_1 - \mu_2 \neq 0$
2. Set $\alpha = 0.05$.
3. The test statistic and its sampling distribution is

$$t_{STAT} = \frac{\bar{D} - 0}{S_D/\sqrt{n}} \sim t_{(n-1)}.$$

4. Calculate the test statistic

$$t^* = \frac{\bar{d} - 0}{s_d/\sqrt{n}} = \frac{0.48 - 0}{0.0837/\sqrt{5}} = 12.8$$

5. Rejection region: $\{t : t > 2.776 \text{ or } t < -2.776\}$
6. Because t^* falls in the rejection region, we reject H_0 . We conclude that there seems to be a difference in the average tire wear for the two types of tires.

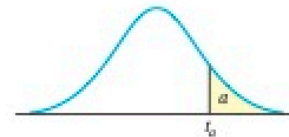


p-VALUE APPROACH

5. The *p*-value is approximated by

$$p\text{-value} = 2P(t_4 > 12.8) < 2 \times 0.005 = 0.01$$

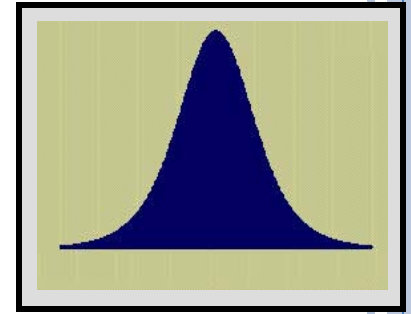
6. Because the *p*-value is less than $\alpha = 0.05$, we reject H_0 . We conclude that there seems to be a difference in the average tire wear for the two types of tires.



■ Table 4 Critical Values of *t*

<i>df</i>	$t_{.100}$	$t_{.050}$	$t_{.025}$	$t_{.010}$	$t_{.005}$	<i>df</i>
1	3.078	6.314	12.706	31.821	63.657	1
2	1.886	2.920	4.303	6.965	9.925	2
3	1.638	2.353	3.182	4.541	5.841	3
4	1.533	2.132	2.776	3.747	4.604	4
5	1.476	2.015	2.571	3.365	4.032	5

SOME NOTES



- Once you have designed the experiment by pairing, you **MUST** analyze it as a paired experiment. If the experiment is not designed as a paired experiment in advance, do not use this procedure.





10.5 INFERENCES CONCERNING A POPULATION VARIANCE

KEY INGREDIENTS

1. **Model:** $X_i \stackrel{i.i.d.}{\sim} N(\mu, \sigma^2)$ for $i = 1, \dots, n$.
2. **Point estimator of σ^2 :** use sample variance S^2 to estimate σ^2 .

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i, \quad S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2.$$

3. **Standardize the point estimator and find its distribution:**

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi_{(n-1)}^2.$$

An idea but not proof: $\frac{(n-1)S^2}{\sigma^2} = \sum \left(\frac{X_i - \bar{X}}{\sigma} \right)^2 \approx \sum Z_i^2$



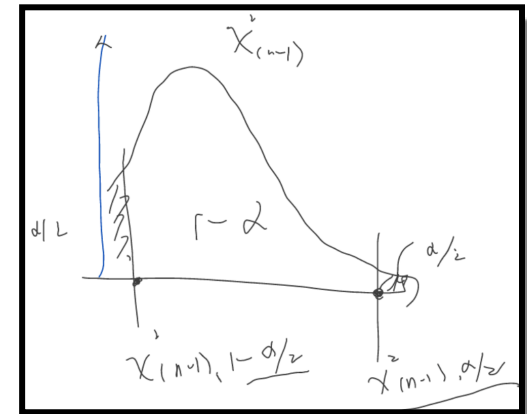
A $100(1 - \alpha) \%$ CONFIDENCE INTERVAL FOR σ^2

- In a similar manner,

$$\begin{aligned}
 1 - \alpha &= P\left(\chi_{(n-1);1-\alpha/2}^2 < \frac{(n-1)S^2}{\sigma^2} < \chi_{(n-1);\alpha/2}^2\right) \\
 &= P\left(\frac{1}{\chi_{(n-1);\alpha/2}^2} < \frac{\sigma^2}{(n-1)S^2} < \frac{1}{\chi_{(n-1);1-\alpha/2}^2}\right) \\
 &= P\left(\frac{(n-1)S^2}{\chi_{(n-1);\alpha/2}^2} < \sigma^2 < \frac{(n-1)S^2}{\chi_{(n-1);1-\alpha/2}^2}\right).
 \end{aligned}$$

- A $100(1 - \alpha) \%$ CI for σ^2 is

$$\left[\frac{(n-1)s^2}{\chi_{(n-1);\alpha/2}^2}, \frac{(n-1)s^2}{\chi_{(n-1);1-\alpha/2}^2} \right].$$



HYPOTHESIS TEST

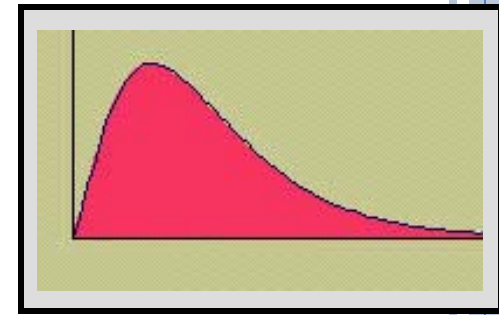
1. $H_0 : \sigma^2 = \sigma_0^2$ v.s. $H_1 : \sigma^2 \neq (> , <) \sigma_0^2$.
2. Set α .
3. Test statistic and its sampling distribution

$$\chi_{STAT}^2 = \frac{(n-1)S^2}{\sigma_0^2} \sim \chi_{(n-1)}^2.$$

4. Calculate the realized statistic, $\chi^{2*} = \frac{(n-1)s^2}{\sigma_0^2}$.
5. Find the rejection region or p -value.
6. Conclude.

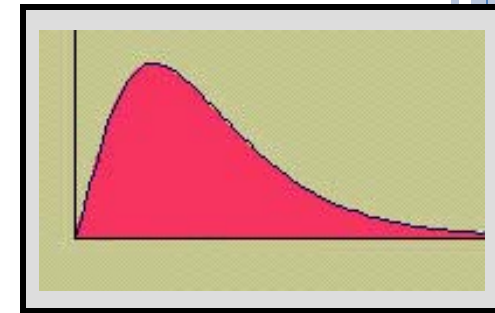


INFERENCE CONCERNING A POPULATION VARIANCE



- Sometimes the primary parameter of interest is not the population mean μ but rather the population variance σ^2 . We choose a random sample of size n from a normal distribution.
- The sample variance s^2 can be used in its standardized form:
$$\chi^2 = \frac{(n-1)s^2}{\sigma^2}$$
- which has a Chi-Square distribution with $n - 1$ degrees of freedom.

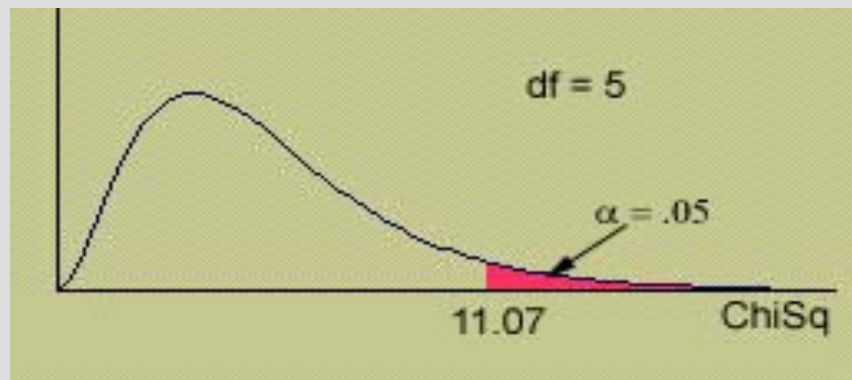
INFERENCE CONCERNING A POPULATION VARIANCE



- Table 5 gives both upper and lower critical values of the chi-square statistic for a given df .

TABLE 5
(continued)

$\chi^2_{.100}$	$\chi^2_{.050}$	$\chi^2_{.025}$	$\chi^2_{.010}$	$\chi^2_{.005}$	df
2.70554	3.84146	5.02389	6.63490	7.87944	1
4.60517	5.99147	7.37776	9.21034	10.5966	2
6.25139	7.81473	9.34840	11.3449	12.8381	3
7.77944	9.48773	11.1433	13.2767	14.8602	4
9.23635	11.0705	12.8325	15.0863	16.7496	5
10.6446	12.5916	14.4494	16.8119	18.5476	6
12.0170	14.0671	16.0128	18.4752	20.0902	7



For example, the value of chi-square that cuts off .05 in the upper tail of the distribution with $df = 5$ is $\chi^2 = 11.07$.

EXAMPLE



•A cement manufacturer claims that his cement has a compressive strength with a standard deviation of 10 kg/cm² or less. A sample of $n = 10$ measurements produced a mean and standard deviation of 312 and 13.96, respectively.

A test of hypothesis:

$H_0: \sigma^2 = 10$ (claim is correct)

$H_a: \sigma^2 > 10$ (claim is wrong)

uses the test statistic:

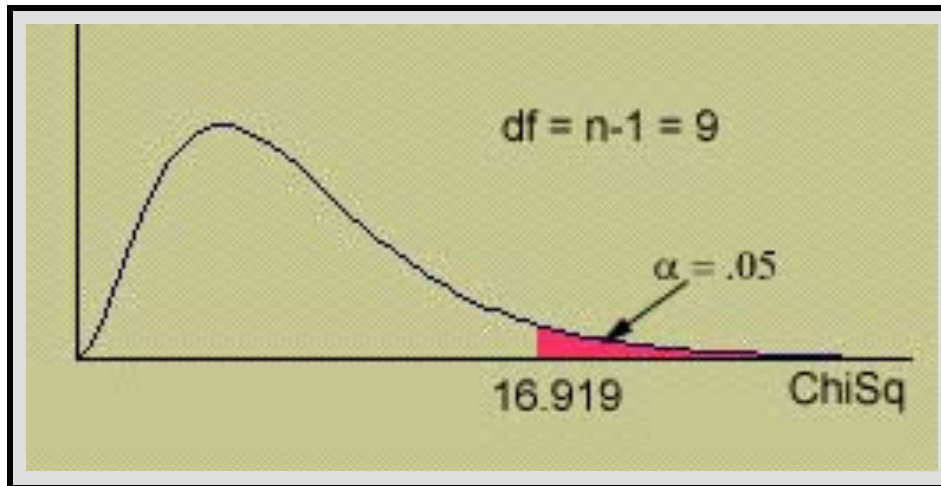
$$\chi^2 = \frac{(n-1)s^2}{10^2} = \frac{9(13.96^2)}{100} = 17.5$$



EXAMPLE



- Do these data produce sufficient evidence to reject the manufacturer's claim? Use $\alpha = .05$.



Rejection region:

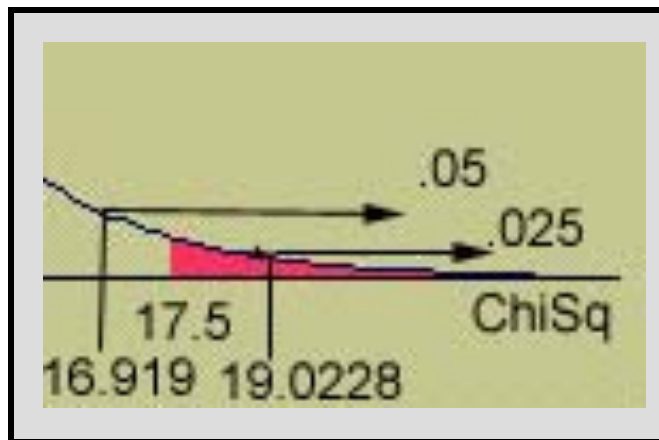
Reject H_0 if $\chi^2 > 16.919$
($\alpha = .05$).

Conclusion: Since $\chi^2 = 17.5$, H_0 is rejected. The standard deviation of the cement strengths is more than 10.

APPROXIMATING THE P-VALUE



p - value : $P(\chi^2 > 17.5)$ with $df = n - 1 = 9$



$.025 < p\text{-value} < .05$

Since the p -value is less than $\alpha = .05$, H_0 is rejected. There is sufficient evidence to reject the manufacturer's claim.

$\chi^2_{.100}$	$\chi^2_{.050}$	$\chi^2_{.025}$	$\chi^2_{.010}$	$\chi^2_{.005}$	df
2.70554	3.84146	5.02389	6.63490	7.87944	1
4.60517	5.99147	7.37776	9.21034	10.5966	2
6.25139	7.81473	9.34840	11.3449	12.8381	3
7.77944	9.48773	11.1433	13.2767	14.8602	4
9.23635	11.0705	12.8325	15.0863	16.7496	5
10.6446	12.5916	14.4494	16.8119	18.5476	6
12.0170	14.0671	16.0128	18.4753	20.2777	7
13.3616	15.5073	17.5346	20.0902	21.9550	8
14.6837	16.9190	19.0228	21.6660	23.5893	9

The slide features a dark blue background. On the left side, there are several vertical stripes of varying shades of blue and white. Overlaid on these stripes are several circles of different sizes, also in shades of blue. The text is positioned to the right of these decorative elements.

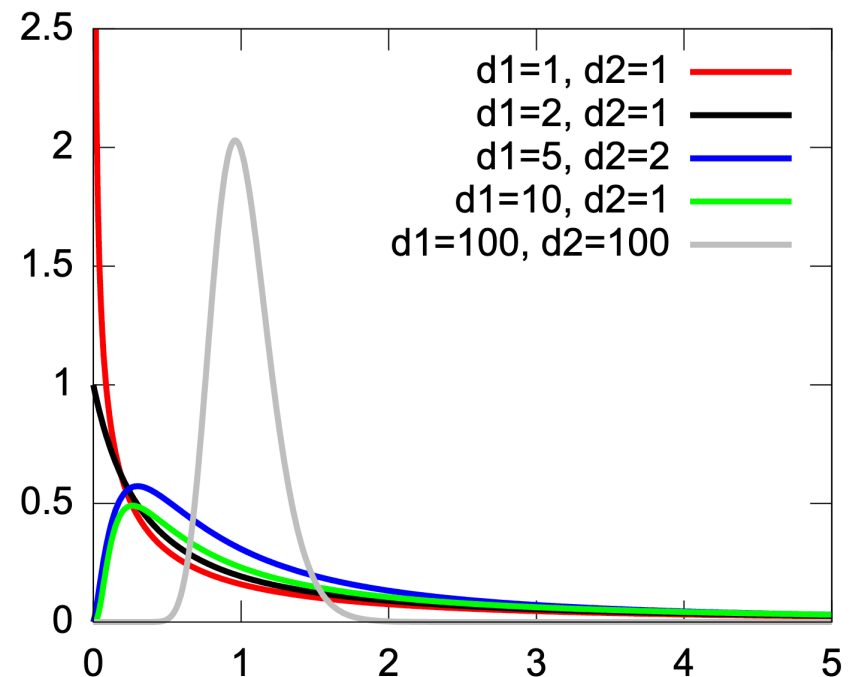
STAT 10.6

Comparing two population variances

THE F DISTRIBUTION

- If $U \sim \chi^2_{\nu_1}$ and $V \sim \chi^2_{\nu_2}$, and they are independent. Then,

$$\frac{U/\nu_1}{V/\nu_2} \sim \frac{\chi^2_{\nu_1}/\nu_1}{\chi^2_{\nu_2}/\nu_2} \sim F_{\nu_1, \nu_2}$$



SAMPLE VARIANCES

- $S_1^2 = \frac{1}{n_1 - 1} \sum_{i=1}^{n_1} (X_{1i} - \bar{X}_1)^2, \quad S_2^2 = \frac{1}{n_2 - 1} \sum_{j=1}^{n_2} (X_{2j} - \bar{X}_2)^2$
- $\bar{X}_1 = \frac{1}{n_1} \sum X_{1i}$
- $\bar{X}_2 = \frac{1}{n_2} \sum_{j=1}^{n_2} X_{2j}$



KEY INGREDIENTS

1. Model

$$X_{1i} \sim N(\mu_1, \sigma_1^2), \quad i = 1, \dots, n_1$$

$$X_{2j} \sim N(\mu_2, \sigma_2^2), \quad j = 1, \dots, n_2.$$

2. Point estimator for σ_1^2/σ_2^2 : S_1^2/S_2^2

3. Standardize the estimator and find its distribution:

$$\frac{\frac{(n_1 - 1)S_1^2/\sigma_1^2}{(n_1 - 1)}}{\frac{(n_2 - 1)S_2^2/\sigma_2^2}{(n_2 - 1)}} = \frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2} \sim F_{(n_1-1), (n_2-1)}$$



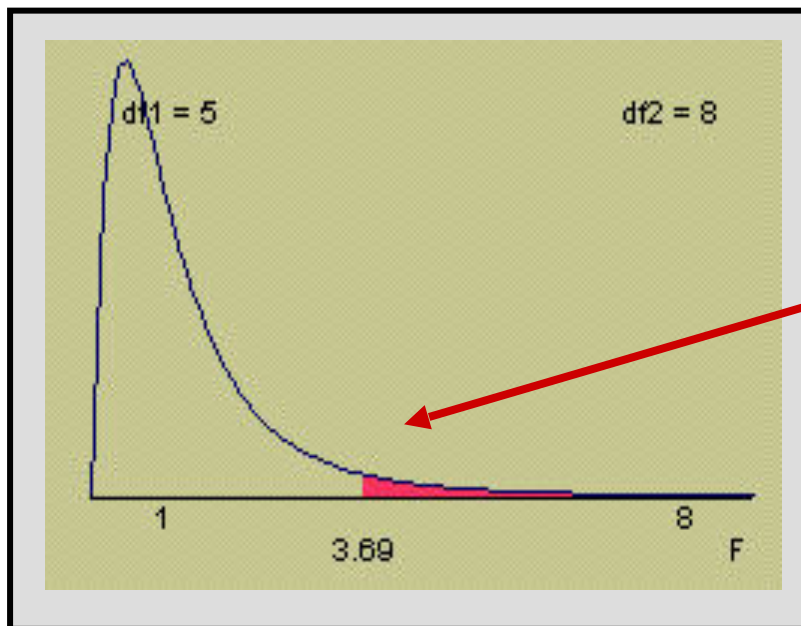
A SMALL INTERMEDIATE STEP

$$\frac{\frac{(n_1 - 1)S_1^2/\sigma_1^2}{(n_1 - 1)}}{\frac{(n_2 - 1)S_2^2/\sigma_2^2}{(n_2 - 1)}} \sim \frac{\chi_{(n_1-1)}^2/(n_1 - 1)}{\chi_{n_2-1}^2/(n_2 - 1)} \sim F_{(n_1-1), (n_2-1)}$$



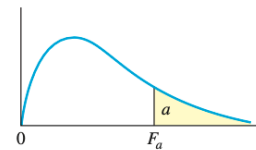
INFERENCE CONCERNING TWO POPULATION VARIANCES

- Table 6 gives only upper critical values of the F statistic for a given pair of df_1 and df_2 .



For example, the value of F that cuts off .05 in the upper tail of the distribution with df_1

■ Table 6 Percentage Points of the F Distribution



df_2	α	df_1								
		1	2	3	4	5	6	7	8	9
1	.100	39.86	49.50	53.59	55.83	57.24	58.20	58.91	59.44	59.86
	.050	161.4	199.5	215.7	224.6	230.2	234.0	236.8	238.9	240.5
	.025	647.8	799.5	864.2	899.6	921.8	937.1	948.2	956.7	963.3
	.010	4052	4999.5	5403	5625	5764	5859	5928	5982	6022
	.005	16211	20000	21615	22500	23056	23437	23715	23925	24091
2	.100	8.53	9.00	9.16	9.24	9.29	9.33	9.35	9.37	9.38
	.050	18.51	19.00	19.16	19.25	19.30	19.33	19.35	19.37	19.38
	.025	38.51	39.00	39.17	39.25	39.30	39.33	39.36	39.37	39.39
	.010	98.50	99.00	99.17	99.25	99.30	99.33	99.36	99.37	99.39
	.005	198.5	199.0	199.2	199.2	199.3	199.3	199.4	199.4	199.4
3	.100	5.54	5.46	5.39	5.34	5.31	5.28	5.27	5.25	5.24
	.050	10.13	9.55	9.28	9.12	9.01	8.94	8.89	8.85	8.81
	.025	17.44	16.04	15.44	15.10	14.88	14.73	14.62	14.54	14.47
	.010	34.12	30.82	29.46	28.71	28.24	27.91	27.64	27.49	27.35
	.005	55.55	49.80	47.47	46.19	45.39	44.84	44.43	44.13	43.88
4	.100	4.54	4.32	4.19	4.11	4.05	4.01	3.98	3.95	3.94
	.050	7.71	6.94	6.59	6.39	6.26	6.16	6.09	6.04	6.00
	.025	12.22	10.65	9.98	9.60	9.36	9.20	9.07	8.98	8.90
	.010	21.20	18.00	16.69	15.98	15.52	15.21	14.98	14.80	14.66
	.005	31.33	26.28	24.26	23.15	22.46	21.97	21.62	21.35	21.14
5	.100	4.06	3.78	3.62	3.52	3.45	3.40	3.37	3.34	3.32
	.050	6.61	5.79	5.41	5.19	5.05	4.95	4.88	4.82	4.77
	.025	10.01	8.43	7.76	7.39	7.15	6.98	6.85	6.76	6.68
	.010	16.26	13.27	12.06	11.39	10.97	10.67	10.46	10.29	10.16
	.005	22.78	18.31	16.53	15.56	14.94	14.51	14.20	13.96	13.77
6	.100	3.78	3.46	3.29	3.18	3.11	3.05	3.01	2.98	2.96
	.050	5.99	5.14	4.76	4.53	4.39	4.28	4.21	4.15	4.10
	.025	8.81	7.26	6.60	6.23	5.99	5.82	5.70	5.60	5.52
	.010	13.75	10.92	9.78	9.15	8.75	8.47	8.26	8.10	7.98
	.005	18.63	14.54	12.92	12.03	11.46	11.07	10.79	10.57	10.39
7	.100	3.59	3.26	3.07	2.96	2.88	2.83	2.78	2.75	2.72
	.050	5.59	4.74	4.35	4.12	3.97	3.87	3.79	3.73	3.68
	.025	8.07	6.54	5.89	5.52	5.29	5.12	4.99	4.90	4.82
	.010	12.25	9.55	8.45	7.85	7.46	7.19	6.99	6.84	6.72
	.005	16.24	12.40	10.88	10.05	9.52	9.16	8.89	8.68	8.51
8	.100	3.46	3.11	2.92	2.81	2.73	2.67	2.62	2.59	2.56
	.050	5.32	4.46	4.07	3.84	3.69	3.58	3.50	3.44	3.39
	.025	7.57	6.06	5.42	5.05	4.82	4.65	4.53	4.43	4.36
	.010	11.26	8.65	7.59	7.01	6.63	6.37	6.18	6.03	5.91
	.005	14.69	11.04	9.60	8.81	8.30	7.95	7.69	7.50	7.34
9	.100	3.36	3.01	2.81	2.69	2.61	2.55	2.51	2.47	2.44
	.050	5.12	4.26	3.86	3.63	3.48	3.37	3.29	3.23	3.18
	.025	7.21	5.71	5.08	4.72	4.48	4.32	4.20	4.10	4.03
	.010	10.56	8.02	6.99	6.42	6.06	5.80	5.61	5.47	5.35
	.005	13.61	10.11	8.72	7.96	7.47	7.13	6.88	6.69	6.54

CI FOR σ_1^2/σ_2^2

- The $100(1 - \alpha)\%$ CI for σ_1^2/σ_2^2

$$\left[\frac{s_1^2}{s_2^2} \frac{1}{F_{\nu_1, \nu_2; \alpha/2}}, \frac{s_1^2}{s_2^2} F_{\nu_2, \nu_1; \alpha/2} \right]$$

- Why?

$$\begin{aligned} 1 - \alpha &= P \left(F_{\nu_1, \nu_2; 1-\alpha/2} < \frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2} < F_{\nu_1, \nu_2; \alpha/2} \right) \\ &= P \left(F_{\nu_1, \nu_2; 1-\alpha/2} < \frac{S_1^2}{S_2^2} \frac{\sigma_2^2}{\sigma_1^2} < F_{\nu_1, \nu_2; \alpha/2} \right) \\ &= P \left(\frac{1}{F_{\nu_1, \nu_2; \alpha/2}} < \frac{1}{S_1^2/S_2^2} \frac{\sigma_1^2}{\sigma_2^2} < \frac{1}{F_{\nu_1, \nu_2; 1-\alpha/2}} \right) \\ &= P \left(\frac{S_1^2}{S_2^2} \frac{1}{F_{\nu_1, \nu_2; \alpha/2}} < \frac{\sigma_1^2}{\sigma_2^2} < \frac{S_1^2}{S_2^2} \frac{1}{F_{\nu_1, \nu_2; 1-\alpha/2}} \right) \\ &= P \left(\frac{S_1^2}{S_2^2} \frac{1}{F_{\nu_1, \nu_2; \alpha/2}} < \frac{\sigma_1^2}{\sigma_2^2} < \frac{S_1^2}{S_2^2} F_{\nu_2, \nu_1; \alpha/2} \right). \end{aligned}$$

WHY $F_{\nu_1, \nu_2; 1-\alpha} = \frac{1}{F_{\nu_2, \nu_1; \alpha}}$?

Note $F_{\nu_1, \nu_2} \sim \frac{\chi_{\nu_1}^2 / \nu_1}{\chi_{\nu_2}^2 / \nu_2} = \frac{1}{\frac{\chi_{\nu_2}^2 / \nu_2}{\chi_{\nu_1}^2 / \nu_1}} \sim \frac{1}{F_{\nu_2, \nu_1}}$

1. First, $\alpha = P(F_{\nu_2, \nu_1} > F_{\nu_2, \nu_1; \alpha})$

2. Second,

$$\alpha = P(F_{\nu_1, \nu_2} < F_{\nu_1, \nu_2; 1-\alpha}) = P\left(\frac{1}{F_{\nu_2, \nu_1}} < F_{\nu_1, \nu_2; 1-\alpha}\right) = P\left(F_{\nu_2, \nu_1} > \frac{1}{F_{\nu_1, \nu_2; 1-\alpha}}\right)$$

Thus, they must be the same.



INFERENCE CONCERNING TWO POPULATION VARIANCES

- We can make inferences about the ratio of two population variances in the form a ratio. We choose two independent random samples of size n_1 and n_2 from normal distributions.
- If the two population variances are equal, the statistic

$$F = \frac{s_1^2}{s_2^2}$$

- has an F distribution with $df_1 = n_1 - 1$ and $df_2 = n_2 - 1$ degrees of freedom.

EXAMPLE



- An experimenter has performed a experiment using two groups of rats. He wants to test $H_0: \mu_1 = \mu_2$, but first he wants to make sure that the population variances are equal.

	Standard (2)	Experimental (1)
Sample size	10	11
Sample mean	13.64	12.42
Sample Std Dev	2.3	5.8

Find the 95% CI for σ_1^2/σ_2^2 !



■ Table 6 (continued)

10	df_1									α	df_2
	12	15	20	24	30	40	60	120	∞		
60.19	60.71	61.22	61.74	62.00	62.26	62.53	62.79	63.06	63.33	.100	1
241.9	243.9	245.9	248.0	249.1	250.1	251.2	252.2	253.3	254.3	.050	
968.6	976.7	984.9	993.1	997.2	1001	1006	1010	1014	1018	.025	
6056	6106	6157	6209	6235	6261	6287	6313	6339	6366	.010	
24224	24426	24630	24836	24940	25044	25148	25253	25359	25465	.005	
9.39	9.41	9.42	9.44	9.45	9.46	9.47	9.47	9.48	9.49	.100	2
19.40	19.41	19.43	19.45	19.45	19.46	19.47	19.48	19.49	19.50	.050	
39.40	39.41	39.43	39.45	39.46	39.46	39.47	39.48	39.49	39.50	.025	
99.40	99.42	99.43	99.45	99.46	99.47	99.47	99.48	99.49	99.50	.010	
199.4	199.4	199.4	199.4	199.5	199.5	199.5	199.5	199.5	199.5	.005	
5.23	5.22	5.20	5.18	5.18	5.17	5.16	5.15	5.14	5.13	.100	3
8.79	8.74	8.70	8.66	8.64	8.62	8.59	8.57	8.55	8.53	.050	
14.42	14.34	14.25	14.17	14.12	14.08	14.04	13.99	13.95	13.90	.025	
27.23	27.05	26.87	26.69	26.60	26.50	26.41	26.32	26.22	26.13	.010	
43.69	43.39	43.08	42.78	42.62	42.47	42.31	42.15	41.99	41.83	.005	
3.92	3.90	3.87	3.84	3.83	3.82	3.80	3.79	3.78	3.76	.100	4
5.96	5.91	5.86	5.80	5.77	5.75	5.72	5.69	5.66	5.63	.050	
8.84	8.75	8.66	8.56	8.51	8.46	8.41	8.36	8.31	8.26	.025	
14.55	14.37	14.20	14.02	13.93	13.84	13.75	13.65	13.56	13.46	.010	
20.97	20.70	20.44	20.17	20.03	19.89	19.75	19.61	19.47	19.32	.005	
3.30	3.27	3.24	3.21	3.19	3.17	3.16	3.14	3.12	3.10	.100	5
4.74	4.68	4.62	4.56	4.53	4.50	4.46	4.43	4.40	4.36	.050	
6.62	6.52	6.43	6.33	6.28	6.23	6.18	6.12	6.07	6.02	.025	
10.05	9.89	9.72	9.55	9.47	9.38	9.29	9.20	9.11	9.02	.010	
13.62	13.38	13.15	12.90	12.78	12.66	12.53	12.40	12.27	12.14	.005	
2.94	2.90	2.87	2.84	2.82	2.80	2.78	2.76	2.74	2.72	.100	6
4.06	4.00	3.94	3.87	3.84	3.81	3.77	3.74	3.70	3.67	.050	
5.46	5.37	5.27	5.17	5.12	5.07	5.01	4.96	4.90	4.85	.025	
7.87	7.72	7.56	7.40	7.31	7.23	7.14	7.06	6.97	6.88	.010	
10.25	10.03	9.81	9.59	9.47	9.36	9.24	9.12	9.00	8.88	.005	
2.70	2.67	2.63	2.59	2.58	2.56	2.54	2.51	2.49	2.47	.100	7
3.64	3.57	3.51	3.44	3.41	3.38	3.34	3.30	3.27	3.23	.050	
4.76	4.67	4.57	4.47	4.42	4.36	4.31	4.25	4.20	4.14	.025	
6.62	6.47	6.31	6.16	6.07	5.99	5.91	5.82	5.74	5.65	.010	
8.38	8.18	7.97	7.75	7.65	7.53	7.42	7.31	7.19	7.08	.005	
2.54	2.50	2.46	2.42	2.40	2.38	2.36	2.34	2.32	2.29	.100	8
3.35	3.28	3.22	3.15	3.12	3.08	3.04	3.01	2.97	2.93	.050	
4.30	4.20	4.10	4.00	3.95	3.89	3.84	3.78	3.73	3.67	.025	
5.81	5.67	5.52	5.36	5.28	5.20	5.12	5.03	4.95	4.86	.010	
7.21	7.01	6.81	6.61	6.50	6.40	6.29	6.18	6.06	5.95	.005	
2.42	2.38	2.34	2.30	2.28	2.25	2.23	2.21	2.18	2.16	.100	9
3.14	3.07	3.01	2.94	2.90	2.86	2.83	2.79	2.75	2.71	.050	
3.96	3.87	3.77	3.67	3.61	3.56	3.51	3.45	3.39	3.33	.025	
5.28	5.11	4.96	4.81	4.73	4.65	4.57	4.48	4.40	4.31	.010	
6.42	6.23	6.03	5.83	5.73	5.62	5.52	5.41	5.30	5.19	.005	

■ Table 6 (continued)

	df_1									
df_2	α	1	2	3	4	5	6	7	8	9
10	.100	3.29	2.92	2.73	2.61	2.52	2.46	2.41	2.38	2.35
	.050	4.96	4.10	3.71	3.48	3.33	3.22	3.14	3.07	3.02
	.025	6.94	5.46	4.83	4.47	4.24	4.07	3.95	3.85	3.78
	.010	10.04	7.56	6.55	5.99	5.64	5.39	5.20	5.06	4.94
	.005	12.83	9.43	8.08	7.34	6.87	6.54	6.30	6.12	5.97

95% CI FOR σ_1^2/σ_2^2

- Check F table:

$$F_{10,9;0.025} = 3.96, \text{ and } F_{9,10;0.025} = 3.78$$

- The 95% CI for σ_1^2/σ_2^2 :

$$\left[\frac{s_1^2}{s_2^2} \frac{1}{F_{\nu_1, \nu_2; \alpha/2}}, \frac{s_1^2}{s_2^2} F_{\nu_2, \nu_1; \alpha/2} \right] = \left[\frac{5.8^2}{2.3^2} \frac{1}{3.96}, \frac{5.8^2}{2.3^2} 3.78 \right] = [1.61, 24.04]$$

- Because the above CI does not overlap 1, the variances should be different.



EXAMPLE



- An experimenter has performed a experiment using two groups of rats. He wants to test $H_0: \mu_1 = \mu_2$, but first he wants to make sure that the population variances are equal.

	Standard (2)	Experimental (1)
Sample size	10	11
Sample mean	13.64	12.42
Sample Std Dev	2.3	5.8

Preliminary test :

$$H_0 : \sigma_1^2 = \sigma_2^2 \text{ versus } H_a : \sigma_1^2 \neq \sigma_2^2$$



HYPOTHESIS TEST

- $H_0 : \frac{\sigma_1^2}{\sigma_2^2} = \frac{\sigma_{1,0}^2}{\sigma_{2,0}^2}$ vs $H_1 : \frac{\sigma_1^2}{\sigma_2^2} \neq \frac{\sigma_{1,0}^2}{\sigma_{2,0}^2}$
- The test statistic and its sampling distribution is

$$F_{STAT} = \frac{S_1^2 / \sigma_{1,0}^2}{S_2^2 / \sigma_{2,0}^2} \sim F_{(n_1-1), (n_2-1)}$$



HYPOTHESIS TEST

1. $H_0 : \frac{\sigma_1^2}{\sigma_2^2} = 1$ vs $H_a : \frac{\sigma_1^2}{\sigma_2^2} \neq 1$

2. Set α

3. Test statistic: $F_{STAT} = \frac{S_1^2}{S_2^2} \sim F_{(n_1-1), (n_2-1)}$

4. Calculate the realized statistic $F^* = \frac{s_1^2}{s_2^2}$

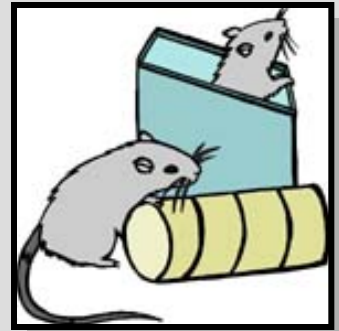
5. Find the rejection region or p-value

6. Conclude

A trick for two-sided test: we designate the sample with the larger standard deviation as sample 1, to force the test statistic into the upper tail of the F distribution!



EXAMPLE



	Standard (2)	Experimental (1)
Sample size	10	11
Sample Std Dev	2.3	5.8

$$H_0 : \sigma_1^2 = \sigma_2^2$$
$$H_a : \sigma_1^2 \neq \sigma_2^2$$

Test statistic :

$$F = \frac{s_1^2}{s_2^2} = \frac{5.8^2}{2.3^2} = 6.36$$

We designate the sample with the larger standard deviation as sample 1, to force the test statistic into the upper tail of the F distribution.

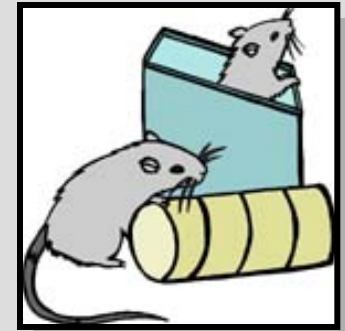


EXAMPLE

$$H_0 : \sigma_1^2 = \sigma_2^2$$
$$H_a : \sigma_1^2 \neq \sigma_2^2$$

Test statistic :

$$F = \frac{s_1^2}{s_2^2} = \frac{5.8^2}{2.3^2} = 6.36$$



The rejection region is two-tailed, with $\alpha = .05$, but we only need to find the upper critical value, which has $\alpha/2 = .025$ to its right.

From Table 6, with $df_1=10$ and $df_2 = 9$, we reject H_0 if $F > 3.96$.

CONCLUSION: Reject H_0 . There is sufficient evidence to indicate that the variances are **unequal**. Do not rely on the assumption of equal variances for your *t test*!

REJECTION REGION APPROACH

1. $H_0 : \frac{\sigma_1^2}{\sigma_2^2} = 1$ vs $H_a : \frac{\sigma_1^2}{\sigma_2^2} \neq 1$

2. $\alpha = 0.05$

3. Test statistic: $F_{STAT} = \frac{S_1^2}{S_2^2} \sim F_{10,9}$

4. Realized test statistic: $F^* = \frac{s_1^2}{s_2^2} = \frac{5.8^2}{2.3^2} = 6.36.$

5. The rejection region: $\{F : F > 3.96\}$ with $F_{10,9;0.025} = 3.96$

Without the trick, the rejection region is $\{F : F < 0.3 \text{ or } F > 3.96\}$,
with $F_{10,9;0.975} = 1/F_{10,9;0.025} = 1/3.78 = 0.3$ and $F_{10,9;0.025} = 3.96.$

6. Because F^* falls into the rejection region, we reject H_0 . We conclude that there is a statistically significant evidence that these two variances are unequal.



■ Table 6 (continued)

df_2	α	df_1								
		1	2	3	4	5	6	7	8	9
10	.100	3.29	2.92	2.73	2.61	2.52	2.46	2.41	2.38	2.35
	.050	4.96	4.10	3.71	3.48	3.33	3.22	3.14	3.07	3.02
	.025	6.94	5.46	4.83	4.47	4.24	4.07	3.95	3.85	3.78
	.010	10.04	7.56	6.55	5.99	5.64	5.39	5.20	5.06	4.94
	.005	12.83	9.43	8.08	7.34	6.87	6.54	6.30	6.12	5.97

■ Table 6 (continued)

df_1										
10	12	15	20	24	30	40	60	120	∞	α
60.19	60.71	61.22	61.74	62.00	62.26	62.53	62.79	63.06	63.33	.100
241.9	243.9	245.9	248.0	249.1	250.1	251.2	252.2	253.3	254.3	.050
968.6	976.7	984.9	993.1	997.2	1001	1006	1010	1014	1018	.025
6056	6106	6157	6209	6235	6261	6287	6313	6339	6366	.010
24224	24426	24630	24836	24940	25044	25148	25253	25359	25465	.005
9.39	9.41	9.42	9.44	9.45	9.46	9.47	9.47	9.48	9.49	.100
19.40	19.41	19.43	19.45	19.45	19.46	19.47	19.48	19.49	19.50	.050
39.40	39.41	39.43	39.45	39.46	39.46	39.47	39.48	39.49	39.50	.025
99.40	99.42	99.43	99.45	99.46	99.47	99.47	99.48	99.49	99.50	.010
199.4	199.4	199.4	199.4	199.5	199.5	199.5	199.5	199.5	199.5	.005
5.23	5.22	5.20	5.18	5.18	5.17	5.16	5.15	5.14	5.13	.100
8.79	8.74	8.70	8.66	8.64	8.62	8.59	8.57	8.55	8.53	.050
14.42	14.34	14.25	14.17	14.12	14.08	14.04	13.99	13.95	13.90	.025
27.23	27.05	26.87	26.69	26.60	26.50	26.41	26.32	26.22	26.13	.010
43.69	43.39	43.08	42.78	42.62	42.47	42.31	42.15	41.99	41.83	.005
3.92	3.90	3.87	3.84	3.83	3.82	3.80	3.79	3.78	3.76	.100
5.96	5.91	5.86	5.80	5.77	5.75	5.72	5.69	5.66	5.63	.050
8.84	8.75	8.66	8.56	8.51	8.46	8.41	8.36	8.31	8.26	.025
14.55	14.37	14.20	14.02	13.93	13.84	13.75	13.65	13.56	13.46	.010
20.97	20.70	20.44	20.17	20.03	19.89	19.75	19.61	19.47	19.32	.005
3.30	3.27	3.24	3.21	3.19	3.17	3.16	3.14	3.12	3.10	.100
4.74	4.68	4.62	4.56	4.53	4.50	4.46	4.43	4.40	4.36	.050
6.62	6.52	6.43	6.33	6.28	6.23	6.18	6.12	6.07	6.02	.025
10.05	9.89	9.72	9.55	9.47	9.38	9.29	9.20	9.11	9.02	.010
13.62	13.38	13.15	12.90	12.78	12.66	12.53	12.40	12.27	12.14	.005
2.94	2.90	2.87	2.84	2.82	2.80	2.78	2.76	2.74	2.72	.100
4.06	4.00	3.94	3.87	3.84	3.81	3.77	3.74	3.70	3.67	.050
5.46	5.37	5.27	5.17	5.12	5.07	5.01	4.96	4.90	4.85	.025
7.87	7.72	7.56	7.40	7.31	7.23	7.14	7.06	6.97	6.88	.010
10.25	10.03	9.81	9.59	9.47	9.36	9.24	9.12	9.00	8.88	.005
2.70	2.67	2.63	2.59	2.58	2.56	2.54	2.51	2.49	2.47	.100
3.64	3.57	3.51	3.44	3.41	3.38	3.34	3.30	3.27	3.23	.050
4.76	4.67	4.57	4.47	4.42	4.36	4.31	4.25	4.20	4.14	.025
6.62	6.47	6.31	6.16	6.07	5.99	5.91	5.82	5.74	5.65	.010
8.38	8.18	7.97	7.75	7.65	7.53	7.42	7.31	7.19	7.08	.005
2.54	2.50	2.46	2.42	2.40	2.38	2.36	2.34	2.32	2.29	.100
3.35	3.28	3.22	3.15	3.12	3.08	3.04	3.01	2.97	2.93	.050
4.30	4.20	4.10	4.00	3.95	3.89	3.84	3.78	3.73	3.67	.025
5.81	5.67	5.52	5.36	5.28	5.20	5.12	5.03	4.95	4.86	.010
7.21	7.01	6.81	6.61	6.50	6.40	6.29	6.18	6.06	5.95	.005
2.42	2.38	2.34	2.30	2.28	2.25	2.23	2.21	2.18	2.16	.100
3.14	3.07	3.01	2.94	2.90	2.86	2.83	2.79	2.75	2.71	.050
3.96	3.87	3.77	3.67	3.61	3.56	3.51	3.45	3.39	3.33	.025
5.26	5.11	4.96	4.81	4.73	4.65	4.57	4.48	4.40	4.31	.010
6.42	6.23	6.03	5.83	5.73	5.62	5.52	5.41	5.30	5.19	.005

P-VALUE APPROACH

5. Approximate the p-value

$$0.01 = 2 \times 0.005 < p\text{-value} < 2 \times 0.01 = 0.02$$

6. Because $p\text{-value} < \alpha$, we reject H_0 . We conclude that there is a statistically significant evidence that these two variances are unequal



MORE...

1. $H_0 : \frac{\sigma_1^2}{\sigma_2^2} = 5$ vs $H_a : \frac{\sigma_1^2}{\sigma_2^2} \neq 5$

2. $\alpha = 0.05$

3. Test statistic: $F_{STAT} = \frac{S_1^2}{S_2^2} \frac{1}{5} \sim F_{10,9}$



The slide features a dark blue background. On the left side, there are several vertical stripes of varying shades of blue and white. Overlaid on these stripes are several circles of different sizes, also in shades of blue. The largest circle is positioned near the top left, with several smaller circles scattered below and to its right.

10.7

Kay concepts

KEY CONCEPTS

Parameter	Test Statistic	Degrees of Freedom
μ	$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$	$n - 1$
$\mu_1 - \mu_2$ (equal variances)	$t = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{s^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$	$n_1 + n_2 - 2$
$\mu_1 - \mu_2$ (unequal variances)	$t \approx \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$	Satterthwaite's approximation
$\mu_1 - \mu_2$ (paired samples)	$t = \frac{\bar{d} - \mu_d}{s_d/\sqrt{n}}$	$n - 1$
σ^2	$\chi^2 = \frac{(n - 1)s^2}{\sigma_0^2}$	$n - 1$
σ_1^2/σ_2^2	$F = s_1^2/s_2^2$	$n_1 - 1$ and $n_2 - 1$

KEY CONCEPTS

I. Experimental Designs for Small Samples

1. **Single random sample:** The sampled population must be normal.
2. **Two independent random samples:** Both sampled populations must be normal.
 - a. Populations have a common variance σ^2 .
 - b. Populations have different variances
3. **Paired-difference or matched-pairs design:** The samples are not independent.



KEY CONCEPTS

II. Statistical Tests of Significance

1. Based on the t , F , and χ^2 distributions
2. Use the same procedure as in Chapter 9
3. Rejection region—critical values and significance levels: based on the t , F , and χ^2 distributions with the appropriate degrees of freedom
4. Tests of population parameters: a single mean, the difference between two means, a single variance, and the ratio of two variances

III. Small Sample Test Statistics

To test one of the population parameters when the sample sizes are small, use the following test statistics:



KEY CONCEPTS

Parameter	Test Statistic	Degrees of Freedom
μ	$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$	$n - 1$
$\mu_1 - \mu_2$ (equal variances)	$t = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{s^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$	$n_1 + n_2 - 2$
$\mu_1 - \mu_2$ (unequal variances)	$t \approx \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$	Satterthwaite's approximation
$\mu_1 - \mu_2$ (paired samples)	$t = \frac{\bar{d} - \mu_d}{s_d/\sqrt{n}}$	$n - 1$
σ^2	$\chi^2 = \frac{(n - 1)s^2}{\sigma_0^2}$	$n - 1$
σ_1^2/σ_2^2	$F = s_1^2/s_2^2$	$n_1 - 1$ and $n_2 - 1$

GENERAL FRAMEWORK OF HYPOTHESIS TEST

1. Set up H_0 vs H_a ($\neq$ or $>$ or $<$)
2. Set the value of α .
3. The test statistic and its sampling distribution
4. Calculate the realized statistic.
5. Find the rejection region or p -value.
6. Conclude.
 - Reject or do not reject H_0 .
 - Conclude in plain language and get back to the scenario.

